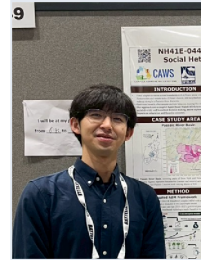


Wenyu Chiou

Ph.D. Candidate · Civil & Environmental Engineering | Complex Adaptive Water Systems (CAWS) Lab,
Lehigh University

My research develops governed LLM-driven agent and multi-agent systems for coupled human–water modeling
— bridging empirical behavioral data, interpretable rule-based simulation, and large language models to study
human–flood interaction and decision-making under risk.

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🌐 github.com/WenyuChiou 🔗 www.linkedin.com/in/wenyu-chiou



§ 01 Education

Ph.D., Civil & Environmental Engineering

Aug 2024 — Present

Lehigh University, United States · Bethlehem, PA, USA

Complex Adaptive Water Systems (CAWS) Lab · Advisor: Prof. Y.C. Ethan Yang · Center of Catastrophe Modeling and Resilience (CMR)

Expected: May 2028

Dissertation: *Advancing Coupled Human-Flood Modeling: From Empirical Foundations to LLM-Driven Multi-Agent Simulation*

Research: LLM-driven coupled ABM–catastrophe model framework for household flood adaptation; governed agent framework (WAGF) enabling LLMs to act-as-humans with behavioral economics grounding; multi-agent systems for quantitative risk assessment.

M.S., Hydrological and Oceanic Sciences

Sep 2021 — Jun 2023

National Central University (NCU), Taiwan · Taoyuan, Taiwan

Surface Hydrology Group · Institute of Hydrological and Oceanic Sciences (IHOS)

GPA 4.3 / 4.3 (max) · Dean's Award, College of Earth Sciences (Jun 2023). Thesis: Investigating groundwater flow characteristics of the Taoyuan coastal zone.

B.S., Earth Sciences

Sep 2017 — Jun 2021

National Central University (NCU), Taiwan · Taoyuan, Taiwan

Department of Earth Sciences

GPA 3.9 / 4.3 (max) · Best Undergraduate Student Project Award (Nov 2020).

§ 02 Research Experience

Ph.D. Candidate — Complex Adaptive Water Systems (CAWS) Lab, Lehigh University

Aug 2024 — Present

- Built a governed LLM-agent framework (WAGF) with validation, execution control, and audit logs — simulates human–environment feedback grounded in empirical survey data.
- Coupled LLM-driven multi-agent system with Catastrophe Flood Model (CAT) for household flood adaptation.
- Implemented a structured agent decision loop with tool use, memory, and reflection for human–environment simulation (Water Resources Research, under review).
- Designed a survey-to-agent behavioral parameterization pipeline (Structural Equation Modeling) — turns empirical data into calibrated agent parameters.

Research Assistant — Surface Hydrology Group, IHOS, National Central University

Aug 2023 — Jun 2024

- Conducted 2D/3D groundwater numerical modeling and data calibration for the Taoyuan coastal zone using HydroGeoChem (HGC).
- Evaluated Nature-Based Solutions (NBS) for groundwater and watershed management, assessing performance and hydrological impacts.

Summer Research Intern — National Science and Technology Center for Disaster Reduction (NCDR)

Jul 2022 — Aug 2022

- Processed and analyzed IPCC AR6 downscaled global climate model ensembles for Taiwan using Python and MATLAB — automated multi-model comparison pipelines across emission scenarios.

Undergraduate Research Intern — Institute of Earth Sciences, Academia Sinica

Jul 2020 — Aug 2020

- Project: “Long-term changes in water cycling and environmental chemical processing.”
- Analyzed hydrogen (δD) and oxygen ($\delta^{18}O$) water isotope data from Feitsui Reservoir.

§ 03 Publications

PEER-REVIEWED & CONFERENCE

- [1] Chiou, W. Y., Yang, Y. C. E., Tanaka, T., Jamrussri, S., & Feng, S. (2026). Household flood adaptation and financial outcomes for homeowners and renters: A coupled agent-based and catastrophe flood modeling framework. *Journal of Hydrology* Under review

§ 04 Awards & Fellowships

Dean's Award, College of Earth Sciences — National Central University	Jun 2023
Interdisciplinary Research Fellowship — Hydrology, Geophysics, Geology — NSTC, Taiwan · \$4,000 USD / year	2021–23
Best Undergraduate Student Project and Report in Earth Sciences — National Central University	Nov 2020
Undergraduate Student Research Project Grant — NSTC, Taiwan · \$1,600 USD total	2020–21

§ 05 Research Interests

- LLM Multi-Agent System
- Agent-Based Modeling
- Catastrophe Modeling
- Human-Water Interaction
- Decision-Making Process
- Structural Equation Modeling

§ 06 Methods & Tools

AGENT SYSTEMS

LLM agents · Multi-agent systems · Structured prompting · Validation rules · Audit logs · Simulation coupling

PROGRAMMING

Python · TypeScript · R · MATLAB · C++

SOFTWARE & TOOLS

Git · Jupyter · SPSS · AMOS · ArcGIS Pro

AI TOOLING

Claude Code · Codex CLI · Gemini CLI · Cursor · Ollama · MCP · Plugin / Skill

§ 07 Open-Source · LLM Agent Systems & AI Research Workflows

WAGF — Water Agent Governance Framework Multi-Agent LLM Governance · 2026 Validates LLM-agent decisions with domain rules, behavioral theory checks, retry feedback, and audit logs before simulation state updates.	ai-research-skills AI research design & workflow skills · Apr 2026 Supports literature review, gap identification, feasibility scoping, manuscript writing, and cross-agent delegation.
awesome-agentic-ai-zh ★ 1.8K+ Community · Education · Apr 2026 Trilingual learning map with 240+ curated resources from LLM basics to multi-agent systems.	agent-collab-skills Multi-Agent System · Orchestration · Apr 2026 Solves token blow-up and silent drift in multi-agent runs — 6–7× avg. token saving (17–22× peak).
codex-delegate AI Research Workflow Tooling · 2025 Routes heavy code to Codex; reasoning stays in Claude — cost-aware multi-model routing.	gemini-delegate-skill AI Research Workflow Tooling · 2025 Delegates large-context synthesis and CJK drafting to Gemini CLI — enables cross-model second-opinion review.

§ 08 Professional Service

Graduate Student Committee Member — Lehigh University, Dept. of Civil & Environmental Engineering	2025 — Present
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§ 09 Community & Open-Source

awesome-agentic-ai-zh Trilingual structured learning curriculum (zh-TW / zh-Hans / EN) — 54 hands-on exercises with dual-path demos (local Ollama + cloud Anthropic), mock-based testing infrastructure, budget-aware framing. 7-stage roadmap from LLM basics to production multi-agent systems. 1,800+ stars · 150+ forks · most-starred repository	Apr 2026
AI Research Skills Marketplace 15 Claude Code skills for the full research workflow (literature → design → manuscript). 5-plugin marketplace, one-command install. Compatible with Codex CLI, Cursor, Windsurf, Hermes. 15 skills · 5 plugins · May 2025 launch	Apr 2026

§ 10 Languages

Mandarin Chinese — Native	English — Professional
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